

```
In [1]: d={}
        type(d)
```

```
Out[1]: dict
```

```
In [2]: d1={'name':'bharadwaj', 'age':392, 'city':'kyoto'}
        d1
```

```
Out[2]: {'name': 'bharadwaj', 'age': 392, 'city': 'kyoto'}
```

```
In [3]: d2={1:'name', 2:'age', 3:'city'}
        d2
```

```
Out[3]: {1: 'name', 2: 'age', 3: 'city'}
```

```
In [4]: d3={'name':'vsc', 'gender':'male', 'employeeed':"no"}
        d3
```

```
Out[4]: {'name': 'vsc', 'gender': 'male', 'employeeed': 'no'}
```

```
In [5]: d1.keys()
```

```
Out[5]: dict_keys(['name', 'age', 'city'])
```

```
In [6]: d1.values()
```

```
Out[6]: dict_values(['bharadwaj', 392, 'kyoto'])
```

```
In [8]: d2.keys()
```

```
Out[8]: dict_keys([1, 2, 3])
```

```
In [9]: d2.values()
```

```
Out[9]: dict_values(['name', 'age', 'city'])
```

```
In [10]: d3.keys()
```

```
Out[10]: dict_keys(['name', 'gender', 'employeeed'])
```

```
In [11]: d3.values()
```

```
Out[11]: dict_values(['vsc', 'male', 'no'])
```

```
In [12]: d1.items()
```

```
Out[12]: dict_items([('name', 'bharadwaj'), ('age', 392), ('city', 'kyoto')])
```

```
In [14]: d2.items()
```

```
Out[14]: dict_items([(1, 'name'), (2, 'age'), (3, 'city')])
```

```
In [16]: d3.items()
```

```
Out[16]: dict_items([('name', 'vsc'), ('gender', 'male'), ('employeeed', 'no')])
```

```
In [17]: d1={1:'name', 2:'age', 'City':['tokyo', 'kyoto', 'osaka']}  
d1
```

```
Out[17]: {1: 'name', 2: 'age', 'City': ['tokyo', 'kyoto', 'osaka']}
```

```
In [18]: d1={1:'name', 2:'age', 'City':['tokyo', 'kyoto', 'osaka'], 'Interests':['photogr  
d1
```

```
Out[18]: {1: 'name',  
          2: 'age',  
          'City': ['tokyo', 'kyoto', 'osaka'],  
          'Interests': ['photography', 'eSports']}
```

```
In [19]: d1={1:'name', 2:'age', 'City':{'tokyo':'japan', 'kyoto':'japan', 'osaka':3}, 'In  
d1
```

```
Out[19]: {1: 'name',  
          2: 'age',  
          'City': {'tokyo': 'japan', 'kyoto': 'japan', 'osaka': 3},  
          'Interests': ['photography', 'eSports']}
```

```
In [21]: keys={'country', 'city', 'age'}  
d4=dict.fromkeys(keys)  
d4
```

```
Out[21]: {'city': None, 'country': None, 'age': None}
```

```
In [26]: keys={'country', 'city', 'age'}  
values = ['japan', 'kyoto', 23]  
d4=dict.fromkeys(keys, values)  
d4
```

```
Out[26]: {'city': ['japan', 'kyoto', 23],  
          'country': ['japan', 'kyoto', 23],  
          'age': ['japan', 'kyoto', 23]}
```

```
In [27]: values.append('yokoso')  
d4
```

```
Out[27]: {'city': ['japan', 'kyoto', 23, 'yokoso'],  
          'country': ['japan', 'kyoto', 23, 'yokoso'],  
          'age': ['japan', 'kyoto', 23, 'yokoso']}
```

```
In [28]: d2
```

```
Out[28]: {1: 'name', 2: 'age', 3: 'city'}
```

```
In [29]: d2[2]
```

```
Out[29]: 'age'
```

```
In [31]: d2.get(1)
```

```
Out[31]: 'name'
```

```
In [32]: d4
```

```
Out[32]: {'city': ['japan', 'kyoto', 23, 'yokoso'],  
         'country': ['japan', 'kyoto', 23, 'yokoso'],  
         'age': ['japan', 'kyoto', 23, 'yokoso']}
```

```
In [33]: d2={'name':'bharadwaj', 'age':34, 'dob':1999, 'city':'kyoto'}  
d2
```

```
Out[33]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999, 'city': 'kyoto'}
```

```
In [34]: d2['name']
```

```
Out[34]: 'bharadwaj'
```

```
In [35]: d2['dob']
```

```
Out[35]: 1999
```

```
In [36]: d2.get('dob')
```

```
Out[36]: 1999
```

```
In [37]: d2
```

```
Out[37]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999, 'city': 'kyoto'}
```

```
In [38]: update = {'country':'japan'}  
d2.update(update)
```

```
In [39]: d2
```

```
Out[39]: {'name': 'bharadwaj',  
         'age': 34,  
         'dob': 1999,  
         'city': 'kyoto',  
         'country': 'japan'}
```

```
In [40]: update1={'job':'AI Engineer'}  
d2.update(update1)  
d2
```

```
Out[40]: {'name': 'bharadwaj',  
         'age': 34,  
         'dob': 1999,  
         'city': 'kyoto',  
         'country': 'japan',  
         'job': 'AI Engineer'}
```

```
In [41]: d2.pop('job') #removes values passed in the functions  
d2
```

```
Out[41]: {'name': 'bharadwaj',  
         'age': 34,  
         'dob': 1999,  
         'city': 'kyoto',  
         'country': 'japan'}
```

```
In [43]: d2.popitem() #removes a random item from the dictionary  
d2
```

```
Out[43]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999}
```

```
In [44]: del[d2['name']]  
d2
```

```
Out[44]: {'age': 34, 'dob': 1999}
```

```
In [45]: d2.clear()  
d2
```

```
Out[45]: {}
```

```
In [46]: del d2
```

```
In [47]: d2
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[47], line 1  
----> 1 d2  
  
NameError: name 'd2' is not defined
```

```
In [48]: d2={'name':'bharadwaj', 'age':34, 'dob':1999, 'city':'kyoto'}  
d2
```

```
Out[48]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999, 'city': 'kyoto'}
```

```
In [49]: d3=d2.copy()  
d3
```

```
Out[49]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999, 'city': 'kyoto'}
```

```
In [50]: id(d2), id(d3)
```

```
Out[50]: (1660094808384, 1660094760896)
```

```
In [51]: d2['country']='japan'  
d2
```

```
Out[51]: {'name': 'bharadwaj',  
          'age': 34,  
          'dob': 1999,  
          'city': 'kyoto',  
          'country': 'japan'}
```

```
In [52]: d3
```

```
Out[52]: {'name': 'bharadwaj', 'age': 34, 'dob': 1999, 'city': 'kyoto'}
```

```
In [53]: for i in d2:  
          print(i)
```

name
age
dob
city
country

```
In [54]: for i in enumerate(d2):  
         print(i)
```

```
(0, 'name')  
(1, 'age')  
(2, 'dob')  
(3, 'city')  
(4, 'country')
```

```
In [55]: d2
```

```
Out[55]: {'name': 'bharadwaj',  
          'age': 34,  
          'dob': 1999,  
          'city': 'kyoto',  
          'country': 'japan'}
```

```
In [56]: 'japan' in d2
```

```
Out[56]: False
```

```
In [57]: 'country' in d2
```

```
Out[57]: True
```

```
In [58]: all(d2)
```

```
Out[58]: True
```

```
In [59]: any(d2)
```

```
Out[59]: True
```

```
In [ ]:
```